```
%pip install tensorflow

Defaulting to user installation because normal site-packages is not
writeable
Requirement already satisfied: tensorflow in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(2.5.1)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in
c:\users\lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(0.2.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(3.4.0)
Requirement already satisfied: packaging in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from tensorflow) (25.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (6.33.6)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(2.32.5)
Requirement already satisfied: setuptools in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from tensorflow) (80.9.0)
Requirement already satisfied: six>=1.12.0 in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(2.5.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
```

```
(1.84.0)
Requirement already satisfied: keras>=3.12.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(3.15.1)
Requirement already satisfied: numpy>=1.26.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from tensorflow)
(2.5.4)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
tensorflow) (0.6.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
requests<3,>=2.21.0->tensorflow) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from
requests<3,>=2.21.0->tensorflow) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from
requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from
requests<3,>=2.21.0->tensorflow) (2025.8.3)
Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from
astunparse>=1.6.0->tensorflow) (0.48.0)
Requirement already satisfied: rich in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from keras>=3.12.0-
>tensorflow) (15.0.0)
Requirement already satisfied: namex in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from keras>=3.12.0-
>tensorflow) (0.1.0)
Requirement already satisfied: optree in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from keras>=3.12.0-
>tensorflow) (0.20.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from rich-
>keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from rich-
>keras>=3.12.0->tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from markdown-it-py>=2.2.0-
>rich->keras>=3.12.0->tensorflow) (0.1.2)
Note: you may need to restart the kernel to use updated packages.

pip install matplotlib numpy
```

```
Defaulting to user installation because normal site-packages is not
writeable
Requirement already satisfied: matplotlib in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (3.10.6)
Requirement already satisfied: numpy in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (2.5.4)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(25.0)
Requirement already satisfied: pillow>=8 in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\lenovo\
appdata\roaming\python\python313\site-packages (from matplotlib)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\
lenovo\appdata\roaming\python\python313\site-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\appdata\
roaming\python\python313\site-packages (from python-dateutil>=2.7-
>matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```python
#. Loading and preprocessing the image data
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt
import numpy as np
# Load CIFAR-10 dataset
(X_train, y_train), (X_test, y_test) = datasets.cifar10.load_data()
# Convert labels to 1D
y_train = y_train.reshape(-1)
y_test = y_test.reshape(-1)
# Normalize images
X_train = X_train / 255.0

X_test = X_test / 255.0
classes = ["airplane", "automobile", "bird", "cat", "deer",
"dog", "frog", "horse", "ship", "truck"]
```

```
print(X_train.shape)
print(X_test.shape)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-
python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 2050s 12us/step
(50000, 32, 32, 3)
(10000, 32, 32, 3)
```

```
cnn = models.Sequential([
    layers.Input(shape=(32, 32, 3)),
    layers.Conv2D(32, (3, 3), activation="relu"),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation="relu"),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation="relu"),
    layers.Dense(10, activation="softmax")
])

cnn.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

cnn.summary()
```

```
Model: "sequential_1"
```

```
┏━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━┓
┃ Layer (type)                         ┃ Output Shape                ┃ Param #         ┃
┡━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━┩
│ conv2d_2 (Conv2D)                    │ (None, 30, 30, 32)          │ 896             │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ max_pooling2d_2 (MaxPooling2D)       │ (None, 15, 15, 32)          │ 0               │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ conv2d_3 (Conv2D)                    │ (None, 13, 13, 64)          │ 18,496          │
├──────────────────────────────────────┼─────────────────────────────┼─────────────────┤
│ max_pooling2d_3 (MaxPooling2D)       │ (None, 6, 6, 64)            │
```

```
0 │
├──────────────────────────────────────┼─────────────────────────────┼
──────────────────┤
│ flatten_1 (Flatten)                  │ (None, 2304)                │
0 │
├──────────────────────────────────────┼─────────────────────────────┼
──────────────────┤
│ dense_2 (Dense)                      │ (None, 64)                  │
147,520 │
├──────────────────────────────────────┼─────────────────────────────┼
──────────────────┤
│ dense_3 (Dense)                      │ (None, 10)                  │
650 │
└──────────────────────────────────────┴─────────────────────────────┴
──────────────────┘

 Total params: 167,562 (654.54 KB)

 Trainable params: 167,562 (654.54 KB)

 Non-trainable params: 0 (0.00 B)
```

```
# c. Training the model
history = cnn.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32
)
```

```
Epoch 1/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 27s 17ms/step - accuracy: 0.7881 -
loss: 0.6074
Epoch 2/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 28s 18ms/step - accuracy: 0.8011 -
loss: 0.5704
Epoch 3/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 25s 16ms/step - accuracy: 0.8086 -
loss: 0.5443
Epoch 4/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 0.8172 -
loss: 0.5181
Epoch 5/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 0.8272 -
loss: 0.4922
Epoch 6/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 16ms/step - accuracy: 0.8359 -
loss: 0.4639
Epoch 7/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 0.8447 -
```

```
loss: 0.4417
Epoch 8/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 27s 17ms/step - accuracy: 0.8512 -
loss: 0.4194
Epoch 9/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 27s 17ms/step - accuracy: 0.8599 -
loss: 0.3955
Epoch 10/10
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 0.8658 -
loss: 0.3782

#d. Estimating the model's performance
loss, accuracy = cnn.evaluate(X_test, y_test)

print("Test Loss:", loss)
print("Test Accuracy:", accuracy)

313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.6827 - loss:
1.2296
Test Loss: 1.2295582294464111
Test Accuracy: 0.682699978351593

#Prediction
y_pred = cnn.predict(X_test)
y_classes = np.argmax(y_pred, axis=1)
print("Actual:", classes[y_test[3]])
print("Predicted:", classes[y_classes[3]])

313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step
Actual: airplane
Predicted: airplane
```